When Reasoning Leaves the Transcript

A Research Program for Hybrid Computation, Semantic Boundaries, and Consequential Agency

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Abstract

In *Hybrid Reasoning Agents*, Jiaru Zou, Yunzhe Qi, Sirui Chen, Federico Bianchi, Shang Zhu, Pan Lu, Jingrui He, and James Zou describe agents that combine text-based interaction with internal computation in continuous representations. This separation raises an engineering question: what must cross from private computation into the world for another participant to interpret a proposal, assess its evidence, determine its authority, and inherit its remaining obligations? This research-program essay develops a recipient- and task-relative account of boundary sufficiency. Its central test constructs source situations that yield the same outward record but legitimately require different decisions, and asks whether an interface preserves the relevant distinction, requests more information, or justifiably refuses the handoff. Five proposed investigations examine representation placement under whole-loop budgets, semantic sufficiency, continuity across model and authority changes, evaluator economics, and cross-agent or institutional interpretation. Earlier architectural and engineering work supplies specific distinctions and experimental starting points. The program separates internal efficiency from evidential warrant, permission, observed effect, and continuing responsibility; its proposed studies haven't been performed.

Keywords: hybrid reasoning, latent reasoning, semantic interfaces, consequential AI agents, evidence and authorization, institutional memory, evaluator economics

A change in what an agent's words mean

A capable engineer can stare at a design, consider a dozen possibilities, recognize an uncomfortable pattern, and finally say, "Check the reset sequence." The listener gets five words. The engineer may have drawn on years of experience, discarded alternatives, and anticipated failures that never became sentences. The usefulness of the answer doesn't depend on receiving a transcript of the engineer's thought. It depends on whether the recommendation is appropriate, whether its meaning survives the handoff, and what happens when somebody acts on it.

Language models have briefly made that distinction less obvious. Text-based agent architectures place instructions, intermediate reasoning, tool calls, explanations, and final answers in the same stream. This arrangement is extraordinarily convenient. It gives software developers a universal integration format and gives observers a trace they can read. It also makes the system pay for expressing internal computation in words, including computation that no external participant actually needs to read.

Zou and colleagues' *Hybrid Reasoning Agents* makes that tradeoff concrete. Their HybridAgent moves between text generation and recurrent computation in continuous latent representations through a lightweight HybridLink. The model still uses text to invoke tools and communicate. Its working memory retains a mixed history of text, latent states, and tool observations. The authors report improved performance and substantial inference savings on their evaluated tasks, using post-training to teach the agent when to enter and leave latent mode [1]. This is an architectural result, not just a cheaper way to print a chain of thought.

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There's a useful caution inside the paper itself. In a MATH500 intervention with the 14-billion-parameter model, accuracy was 88.2% with the original latent states, 87.2% after replacing them with nearest-token embeddings, and 44.6% after random-token replacement. The latent sequence carries relevant information, but this test doesn't establish a wholly independent, nonlinguistic form of cognition. The authors also report that visible text traces remain useful for several monitoring and debugging tests. Useful monitoring isn't proof that the visible account faithfully reproduces all computation that influenced an answer [1].

Once internal computation, outward communication, and consequential action are allowed to use different representations, system designers must decide what the external interface is for. A readable account isn't automatically a reliable account; a correct account isn't automatically adequate evidence; and good evidence isn't permission to act. An interface that preserves all the words but loses one relevant condition may be worse than a short, structured record that preserves the conditions on which the next decision depends.

This essay lays out a research program centered on that problem. What must cross from an agent's private computation into the world so that a recipient can interpret the proposal correctly, judge what the evidence supports, determine what's authorized, and inherit any remaining responsibilities? These are related but different tests. Their answers depend on the recipient's task, the conditions in the world, and the possible consequences. A representation may be sufficient for monitoring a machine and insufficient for commanding it. My aim isn't to discover one universal language of responsible AI. It's to engineer boundaries that preserve the distinctions each legitimate decision requires and recognize when they can't.

The discussion that follows connects hybrid reasoning with my earlier papers on control, representation, authority, and institutional memory. These connections help turn the interface problem into specific engineering questions and proposed experiments.

Computation is free to move; authority is not

In my earlier paper *The Dynamic Control Plane* [2], I distinguished a system's capabilities from the control machinery that selects, coordinates, and times their use. *The Capability Mobilization Gap* [3] developed that distinction into a diagnosis problem: a capability can be present and demonstrable, yet fail to appear when it's needed. Hybrid reasoning supplies a new decision for such a controller. At a given moment, should the agent continue its internal latent sequence, formulate explicit text, call a tool, ask for external evidence, request human judgment, or stop?

Those choices aren't just presentation choices. Each carries a different cost, latency, failure mode, and opportunity to learn from the world. The natural extension is a control policy that allocates computation across representations and external resources according to the problem at hand. A system could learn that some questions justify several internal steps before a tool call, that others require early measurement, and that still others can't be answered until an ambiguity has been put to a human. It could discover that a concise representation is sufficient for one recipient and dangerous for another.

In *Reversible Control Placement* [4], I asked when a control function could be moved from an explicit outer layer into an agent or inference runtime without losing its independently held specification, authority, evidence requirements, and executable fallback. The distinction becomes especially important if an agent's internal decisions are conducted in continuous states. Moving the computation that selects an action inward may be beneficial. Moving the authority to decide whether that action is permitted inward is a separate decision, with a different burden of proof.

Imagine a factory maintenance agent that predicts a machine failure. It may legitimately choose diagnostic measurements, reorder reversible checks, and estimate competing causes. A reliable prediction could support a proposal to slow the machine. A calibrated sensor reading could provide evidence that the specified shutdown condition has been met. A designated operator or standing policy might then authorize the intervention.

Finally, an independent observation must establish what the intervention actually did. These aren't four ways of reporting the same confidence level. They're separate achievements with separate failure modes. The system must not let improved internal prediction silently rewrite the evidence requirements, permission rules, or responsibility for the outcome.

This is also a representation problem. In *What Language Should an Agent Act In?* [5], I asked how the agent-facing representation changes the rate of independently accepted work under a fixed task, tool set, acceptance contract, risk policy, and whole-loop budget. The language in which an agent acts isn't the language in which it reasons. But the same discipline applies: let a representation earn its place through accepted outcomes and bounded risk, not because its syntax is familiar or its token count is low. An agent may reason in latent states, propose a structured design change, and submit that change to formal verification. Each stage deserves a representation suited to its purpose.

What the crossing must preserve

An interface is commonly described as a place where bits, strings, structures, or API calls cross. Those items can arrive intact without the recipient understanding what matters. Meaning isn't just a payload sealed inside a message; it depends on the claim or function the source presents, what the recipient is being asked to do, the evidence connecting the message to the world, and the practices that determine correct use. A field containing a design constraint may be syntactically valid and still omit an exception that mattered to the engineer who supplied it. A legal recommendation may arrive at an engineering tool without its conditions. A probability may become a yes-or-no instruction. A human expert may understand why a proposal is unsafe and have no way to express the objection inside a rigid schema. The failure is in the relationship established by the interface, not necessarily in its ability to transmit data.

These failures aren't necessarily failures of either component. An optimizer can solve precisely the problem it received. A policy engine can enforce exactly the rule it was given. A language model can produce an elegant explanation. Yet the complete arrangement can still be wrong if an assumption was added, a qualification vanished, or the receiving party interpreted the message as authorizing more than the sender established. Correctness belongs to a specified contract; adequacy belongs to a contract in use. An engineering investigation should be able to identify whether the source problem was misunderstood, the local task was performed incorrectly, or the output was interpreted beyond its warrant.

When a hybrid model performs useful work without emitting every step as text, a later verbal account is an interpretation of its internal state, not an authenticated recording of a textual conversation that actually took place. The nearest-token probe in HybridAgent is interesting because it makes latent computation partly legible. It doesn't certify that all relevant distinctions, doubts, dependencies, or incentives survive in the resulting pseudo-text [1]. Fluency and fidelity are different properties.

For an important crossing, the system should distinguish failures that look similar from the outside. Did the inbound interpretation misstate the original problem? Did the component fail to satisfy the contract it actually received? Did the outbound interpretation overstate its result? Was a correctly conveyed assertion unsupported by adequate evidence? Was an adequately supported proposal outside the recipient's authority? Did an apparently completed transaction leave somebody else with an unrecognized duty? Each question suggests a different repair. Combining them into an undifferentiated confidence score would conceal the very distinctions the boundary is meant to protect.

A consequential crossing carries several kinds of content that must not be confused. The claim says what's asserted about the world. Its warrant identifies the evidence and assumptions that justify relying on it for a specified purpose. Its practical force says whether it's a hypothesis, warning, recommendation, request, permission, or binding instruction. Its continuing obligations say what must be checked, preserved, revisited,

or handed off after the immediate exchange. A recipient can understand a claim perfectly and still lack good grounds to accept it. It can possess good evidence without having permission to act. It can carry out a permitted action and still fail to achieve its intended effect. None of these distinctions requires an exhaustive transcript of the sender's internal computation, but each may require its own externally checkable mechanism.

One principle follows: an interface must be allowed to refuse an inadequate translation. Suppose a tool exposes only APPROVE or REJECT, but the actual finding is “approved only if this measurement remains within a specified range and is checked again before execution.” Returning APPROVE silently removes the condition; returning REJECT may prevent work that could safely proceed. The proper response may be to demand a richer contract, preserve the outstanding condition in a separately enforced obligation, or escalate the decision. This capacity for a justified refusal is part of intelligent interpretation. Otherwise, each successful-looking exchange may add to a growing stock of hidden assumptions that future participants can’t recover.

A research program could treat the boundary record as a technical object in its own right: a typed account of the proposed action, the result actually claimed, its evidence and assumptions, unresolved qualifications, applicable authority, and the observations or duties still required. The record doesn’t have to reproduce the computation that created it. Nor should it be judged sufficient just because the recipient can paraphrase it fluently. The record is sufficient only relative to a declared class of decisions and operating conditions, when recipients can make the distinctions needed to decide correctly, including when they must not decide. This is a stronger and more practical notion of sufficiency than faithful compression of a transcript.

Consider two different source situations that produce the same external crossing record. In one, the required evidence is valid; in the other, it has expired. Or in one, a recommendation falls within existing authority; in the other, an exception requires separate approval. If the same message obscures these differences, its recipient may take an action appropriate in one situation and unjustified in the other. By contrast, two agents might arrive at the same well-supported conclusion through entirely different latent computations without creating any difference the recipient needs to see. The defect isn’t loss of access to an internal vector. It’s the loss of a distinction that matters to the receiver's legitimate decision.

To test this account of sufficiency, construct pairs of source situations that yield the same outward message but demand different responses under the declared task, evidence, and rules. The oracle for that difference must be grounded independently of the model's self-report, for example through a testable contract, a withheld measurement, or a stipulated policy condition. Vary the receiving task, assumptions, evidence, and permissions. Ask whether the interface exposes the mismatch, requests additional information, or refuses the crossing. No single record will be adequate for every future use. A dependable system should identify the scope of its own adequacy.

An action is a transition, not a confession

There’s a temptation to respond to hidden internal reasoning by demanding more explanation. Internal observability is valuable for debugging, detecting failure modes, and diagnosing abuse. Some uses may require particular forms of transparency. But an agent can supply a plausible account and still act on false evidence or outside its authority. Evaluators should distinguish an explanation of how the agent describes its choice from grounds that justify accepting a claim, permission to perform an action, and evidence that the action achieved its purpose. Inspectability matters, but the most elaborate transcript can’t take the place of these other tests.

Controlling Consequential Transitions develops a seven-stage execution contract: propose, gather evidence, evaluate, authorize, execute, observe, and commit [6]. The sequence separates different kinds of acts that may have the same subject matter. “This change is likely to help” is a prediction; “the required tests passed” is an

evidence claim; “this change is approved for release” is an act of authorization under an applicable policy. The last statement doesn’t acquire authority because it sounds certain. It acquires its force from the role and rules governing who may issue it. Control belongs at the real effect boundary, where software changes records, deploys code, grants access, spends money, or commands a device. For a hybrid reasoner, this external contract can remain explicit even if proposal generation is largely latent. The agent's interior can become faster without making the outer system guess which institutional act just occurred.

Agents Act. Systems Remember. Rules Authorize. People Govern. adds an important independence condition: an agent seeking new authority must not manufacture all the evidence needed to grant it [7]. A model's own explanation of its latent reasoning can inform a review without simultaneously serving as claim, verification, and authorization. In a reliable arrangement, the evidence source, evaluating function, and authority granting permission can be independent where the stakes warrant it. That independence is substantive, not cosmetic: a signature, credential, or approval field is useful only when its issuer is entitled to make the decision and its scope covers the actual requested effect.

This doesn’t imply that humans approve every tool call. It suggests deliberately placed checkpoints: narrow routine actions may proceed within explicit standing authority; costly, risky, irreversible, or surprising actions require stronger evidence or approval from a different authorized person or agent. An agent shouldn’t have to seek permission to think harder. It may have to seek permission to execute a production change. It may also have to seek a more authoritative evaluator before its own confidence can support the requested commitment.

Staying Capable in a World That Pushes Back broadens the time horizon. An action changes the environment and the conditions for the next task: evidence may become stale, permissions may expire, rollback options may disappear, and duties may accumulate [8]. A statement that was adequate when issued can become misleading when the world changes. A crossing record should identify its conditions of validity and the events that require reconsideration. It should also specify how actual effects will be observed and who inherits unfinished work. The system's responsibility doesn’t end when the originating model has finished generating tokens or when a tool reports that a command succeeded.

A visible trace may let an observer predict whether an answer is correct, as the HybridAgent monitoring results partly suggest [1]. That’s different from determining whether a transaction had sufficient independent evidence, whether authorization covered its exact effects, or whether the resulting state is safe for the next participant. These are separate outcomes. Experiments can test each outcome separately.

Memory that survives the thinker

HybridAgent’s treatment of working memory raises a second challenge. Its mixed text-and-latent trajectory is preserved in the model’s accumulated attention state across turns [1]. Under the paper’s stated conditions for continuing with a fixed model, the scheme retains computational context without translating every latent step back to text. It’s valuable precisely because it carries information that an ordinary transcript would lose.

But a Transformer cache isn’t an institutional memory system. A useful cache may depend on a particular model version, inference implementation, position sequence, and the surrounding computational environment. It can support the next forward pass without supporting transfer to another model, investigation six months later, or recovery after a failed deployment. It may preserve a calculation without preserving who had authority to use it.

From Replayed Context to Persistent Time separated recorded history, working context, and persistent control state [9]. The distinction becomes more consequential when replaying a textual conversation can’t reconstruct the state that produced the agent's decisions. A system may need fast temporary state for active computation, a bounded episode context, and durable records of goals, evidence, permissions, and obligations that survive changes of model and operator. Yet a durable record isn’t automatically a durable truth. It must preserve

provenance, applicable scope, invalidation conditions, and mechanisms for revision. Good institutional memory helps a successor identify what remains supported, what has expired, and what needs to be tested again. Keeping everything forever wouldn't solve that problem.

The Clocks Are a Society: Memory Is Duty imagined transient computational participants occupying more durable offices [10]. The useful obligation isn't to preserve every passing internal state indefinitely. It's to preserve responsibilities and the authority to revise them across succession. The agent carrying a task today may not be the agent carrying it tomorrow. If the first agent can hand off only an opaque cache, the next may inherit impressive computation with no reliable way to know which commitments remain binding.

That suggests a pair of distinct engineering services. Computational continuity tries to resume an internal state faithfully when the same compatible system continues. Institutional continuity reconstructs the task, its evidence, permissions, and duties in a form a different participant can use. A good agent architecture might have both. It might retain a model-specific checkpoint to avoid unnecessary recomputation and also emit durable crossing records at selected moments. The latter are intended to survive a change of model or operator. They're not a textual replacement for every hidden calculation; they're an independently usable description of what the ongoing work owes the world.

Intelligence by Appointment adds the notion of temporary, revocable access to capability and resources [11]. Applied here, a latent reasoning session might hold an appointment to use a particular model, private context, evaluator, and computation budget. The appointment could expire, be reassigned, or be revoked without erasing the institution's standing obligations. This becomes a practical test of the separation between private cognitive machinery and continuing responsibility.

One experiment would interrupt an agent immediately before and immediately after a consequential crossing. First resume the same model from a computational checkpoint. Then attempt the task with an updated model, an unrelated model, and a human operator given only the durable record. Which commitments survive? Which permissions are still interpretable? Which facts are incorrectly assumed to be current? Which steps must be repeated? If the system can't answer these questions, it has preserved activity without preserving dependable continuity.

What gets expensive when thinking gets cheap?

The cost of internal reasoning matters because it changes what an agent can afford to consider. If candidate hypotheses and preliminary plans become cheaper, more candidates can be generated before the system commits to a physical experiment, a formal check, or an expensive deployment. That may be excellent news. It may also move the economic pressure to a different part of the workflow.

When Design Exploration Gets Cheap frames this as evaluator economics: choosing what evidence to obtain next, with which evaluator, at what cost and latency, and with what level of fidelity and authority [12]. Hybrid reasoning is one possible reason the upstream side of that equation may change. It doesn't make the external measurements cheaper by the same proportion. The model's reported decoding savings aren't an estimate of the cost of a semiconductor design iteration or an industrial investigation.

Logic Optimization Is Becoming a Learning System treats a candidate circuit as an evidence-carrying counterfactual, connected to assumptions, contract, validation state, cost, and history [13]. This is a promising home for the proposed research. Imagine an optimization agent that can investigate hundreds of possibilities cheaply, but can afford only a limited number of timing runs, formal equivalence checks, placement experiments, or silicon-informed measurements. The important result isn't how many candidates the agent considered. It's how effectively it routed scarce evaluation toward candidates that ultimately survived independent checks.

The Third Clock offers a related view from longer hardware development cycles [14]. Learning happens within individual trials and across the apparatus that decides which trials to run, what to observe, when to seek stronger evidence, and when commitments can safely be made. If hybrid reasoning changes the number of proposals per unit time, this slower learning machinery may become the decisive economic actor. A cheap speculative idea is useful only if the organization has a way to learn from its encounter with reality.

There's a counterargument worth taking seriously. Better internal reasoning might decrease the demand for external evaluation by eliminating weak candidates or producing reliable internal predictions. In some regimes the best result could be fewer physical experiments, not more expensive evaluator capacity. The research shouldn't assume in advance that cheap reasoning necessarily raises evaluator spending or profit. It should ask when it increases the marginal value of external evidence, when it reduces that demand, and who captures the savings.

An evaluator may hold valuable information it can't disclose. *External Analysis of Protected Data* describes existing controlled computation methods, data clean rooms, confidential computing, approved queries, and their limits [15]. A fab, hospital, or industrial operator might possess measurements that are unusually valuable for validating another party's proposed model. The owner could allow bounded evaluation against those measurements without releasing the records. The result itself would still need a disclosure policy: repeated queries, aggregates, model updates, and metadata can leak information even when raw data never leaves the protected environment.

One possible service would sell qualified answers to approved questions without exposing the source records. An outside agent submits a candidate; an owner-controlled evaluator returns a result with validity conditions and a disclosure policy. Economic value might accrue to the repeatable evaluation capability and the evidence behind it, not to the generic model that proposed the candidate. This is an economic hypothesis, not an observed market. Its engineering test is whether the recipient receives enough information to act on the result without learning or claiming more than the evaluator is authorized to release.

Here the research questions of computation, authority, and economics meet. The same boundary that keeps a model from acting on unverified conclusions may also let a data owner offer useful evaluation without surrendering its information capital. But a successful transaction requires four different things: useful computation, technically credible evidence, authorized disclosure, and agreement about what future learning each party may retain. Treating these as one binary "trusted" label would hide the most important design choices.

Interpreters between worlds

The authors of *Hybrid Reasoning Agents* suggest a future in which one agent's hybrid computational trajectory could become a message to another [1]. It's easy to imagine the attraction. Why repeatedly compress rich internal work into sentences if two compatible systems can exchange the richer representation directly? The gain could exceed a reduction in token counts. It might preserve alternatives, relationships, and intermediate structure that ordinary summaries discard.

But compatibility has a price. A latent state meaningful to one model family, checkpoint format, or training history may be useless or actively misleading to another. It may convey protected information without an adequate disclosure check. It may cause the recipient to act on assumptions that were never stated. A transfer that succeeds numerically isn't necessarily a transfer that succeeds semantically.

A likely future has several kinds of crossing, not one universal language. Trusted, compatible agents might exchange internal computational state over a narrow, versioned channel. Other participants might need source code, design representations, numerical arrays, signed measurements, an executable plan, or a structured legal commitment. Humans may need a few sentences that faithfully state the decision and its grounds. The right

representation depends on the relationship between sender and recipient, the nature of the work, and the consequence of misunderstanding it.

The interpreter must preserve the distinctions the receiving participant is entitled or required to use: which conclusions follow from a result, which assumptions support it, what the result leaves undecided, and which actions its institutional role permits. The mediator should distinguish what the source established from what it infers or guesses. Different recipients can legitimately need different interpretations. A monitoring service may need an early warning; a formal checker needs an exact claim and assumptions; an approving operator needs evidence, consequence, authority, and an option to decline. There’s no requirement that all three receive the same description, only that each receive what its own decision requires without being misled about the limits.

The long-term engineering opportunity is to make boundaries progressively better at recognizing both fidelity and inadequacy. A mediator may provide no translated answer at all. It may identify a missing qualification, obsolete policy, incompatible model state, unrecognized expert objection, or target schema that can’t carry the necessary distinction. The system should allow the mediator to challenge the interface and request a different kind of crossing. Successful interpretations may later become deterministic adapters, typed contracts, tests, or narrow models when their conditions have become well understood. But the same hardening can be dangerous if those conditions change unnoticed. A reliable interface needs a way to recognize that yesterday's acceptable translation has stopped being sufficient today.

One program, several tests

The research program should begin with demanding but observable tasks, not an attempt to implement a universal architecture of meaning. A first testbed could use an AI-assisted design optimization workflow with a fixed design contract, candidate-generation tools, formal and implementation evaluators, a limited evaluation budget, and an independent acceptance rule. A second could use a recoverable operational incident with durable state changes, explicit authority rules, and a separate observer of real effects. A protected-data evaluation service would make a useful third setting once the earlier contracts have been tested. Each testbed should make the recipient's role and allowable decisions explicit. The same evidence can be adequate for one recipient and inadequate for another.

These testbeds have different strengths. Hardware design provides precise constraints and costly external checks. Incident response provides time, responsibility, recovery, and the possibility of an action whose visible success hides damage to the next task. Protected-data evaluation adds an owner whose information has value and a requester who must be prevented from learning more than authorized. The same architecture may not win in all three. Differences between them can reveal which boundary properties are intrinsic and which are local to a field.

The first investigation is representational placement. Keep the task information, tools, evaluator access, acceptance contract, and total budget fixed. Hold the foundation model as constant as the methods permit. Compare text-only reasoning, hybrid latent/text reasoning, structured intermediate work, and policies that deliberately decide when to switch modes and call tools. Model adaptation and post-training must be documented so an advantage isn’t falsely attributed to representation alone. Measure independently accepted results, elapsed time, model and tool costs, unsuccessful attempts, repairs, and the information the recipient needed but didn’t receive. Use the accepted-change discipline proposed in *What Language Should an Agent Act In?* [5]. The hypothesis isn’t that latent reasoning will always win, but that a system can allocate representations according to the decisions they support, allowing internal efficiency without sacrificing appropriate external accountability.

The second investigation asks when a crossing record becomes insufficient for a particular recipient and decision. Start with cases where two different source situations are deliberately made to produce the same outward record, yet the evidence or rules call for different next actions. For example, a design may satisfy the same reported timing estimate in two cases, but in only one is the estimate based on calibration valid for the current operating range. If the receiving tool is authorized to proceed only with valid evidence, a record that reports the number and omits its validity conditions is inadequate. It can't support the required distinction. The oracle for correctness comes from the contract and independently established test conditions, not from a private cognitive state.

Construct such pairs systematically. Vary one hidden condition at a time: supporting evidence, source assumptions, uncertainty, validity range, permissions, expected effect, or unfinished obligation. Give independent recipients the crossing record and the external information they're legitimately allowed to see. Test whether they accept the right cases, reject the wrong ones, request missing information, or recognize that their interface can't represent the case at all. Separately measure correct interpretation, justified reliance, valid authorization, and subsequent handling of obligations. False acceptance matters, but unjustified refusal and unnecessary escalation matter too.

Now vary the recipient and purpose. A preliminary diagnostic suggestion may not need an exact formal proof; a release authorization may. The same summary could be adequate for an engineer deciding what to investigate and inadequate for a controller initiating an irreversible action. Success means preserving the distinctions relevant to a declared set of permitted decisions, not reconstructing the original thought process. The strongest counterexample is a pair of situations collapsed into the same message that the recipient must treat differently. A good boundary exposes that collision or refuses the handoff.

The third investigation puts continuity under stress. Interrupt active work after internal computation, after an evaluator call, after authorization, during execution, and after an apparently successful result. Resume from a raw computational checkpoint, a text transcript, and an externally governed record, first with the same model and then with an upgraded model, another model family, and a human participant. Give successors a record containing claims with explicit validity conditions, not timeless declarations of fact. Introduce a changed policy, a revoked permission, and evidence that becomes obsolete between the original action and the handoff. Measure duplicate actions, undetected stale evidence, incorrect permission handling, recovery expense, remaining obligations, and quality of downstream work. A successful system can discard much of its transient cognitive state without discarding the legitimate responsibilities of the work, and it can recognize when yesterday's conclusion needs fresh evidence.

The fourth investigation examines the economics of obtaining justified conclusions. Vary the cost, speed, coverage, independence, and authority of evaluators in the design testbed. Let candidate generators run under different reasoning costs, including a hybrid mode when technically available. Ask whether evaluator-routing policies improve independently accepted design quality under a fixed total budget. Include cheap screening models, expensive higher-fidelity checks, and a withheld reference outcome. Keep evaluator accuracy distinct from the authority of its result: a highly predictive surrogate may help select a candidate without satisfying the evidential standard for release. Test both directions of the economic hypothesis. More cheap proposals might increase the value of external checks; better internal discrimination might reduce the number required. Track evidence retained for future decisions, since a useful rejected candidate may still improve the next search.

The fifth investigation concerns agent-to-agent and agent-to-institution boundaries. Compare an ordinary textual handoff, a typed evidence-and-commitment record, and direct latent-state exchange between compatible models. Give recipients different legitimate roles: a peer reasoning agent, an evaluator, a policy authority, and an operator. Then introduce model-version changes, stale context, incomplete instructions, incompatible assumptions, and protected information the receiver isn't entitled to obtain. Measure task utility, semantic distortion, false conclusions, unjustified acts, leakage, portability, recovery, and the ability to

identify which crossing failed. A direct latent exchange may perform best between trusted compatible agents, yet convey too little portable meaning or too much private context for use as an institutional contract. If a narrow adapter succeeds repeatedly, test whether it can be hardened into an ordinary typed interface and whether it notices when the underlying assumptions have changed.

Across all five investigations, measure what the system leaves behind. An agent might produce a correct answer quickly and still leave unexplained privileges, stale claims, broken recovery paths, or expensive cleanup for others. Keep the main outcomes separate. Semantic fidelity asks whether the recipient understood the relevant distinctions. Evidential warrant asks whether the claim was adequately supported for its use. Authority asks whether the action was permitted under actual rules and roles. Continuity asks whether obligations and validity conditions survived. Cost and latency determine whether the arrangement is worth operating. A single score called “trust” can’t substitute for these tests. Evaluate strong simpler alternatives: fixed-format policies, good retrieval, conventional agents with evaluators, carefully engineered transcripts, and human review where appropriate. Hybrid computation earns its complexity when the whole arrangement performs better across the measures that matter, without hiding a serious regression inside an average.

An experiment could fail in useful ways. If ordinary text records preserve every distinction needed for the tasks and decisions tested, specialized latent-state governance may add little value there. If evaluators can’t agree even with rich information, the problem may lie in the task contract or evidence standard, not its compression. If model changes destroy useful continuity despite durable records, my account of transferable responsibility is incomplete. If better reasoning reduces external evaluation costs, the economic prediction should be revised. If an agent can’t recognize when its output interface is inadequate, that may become the central engineering challenge. These outcomes should refine the research program, not be explained away as implementation inconveniences.

A different kind of intelligence infrastructure

Imagine a future agent performing useful internal computation in representations other participants can’t inspect, cooperating with compatible models over fast numerical channels, acquiring measurements from protected evaluators, and communicating selected results in ordinary language. Its speed would be impressive. The more revealing test is what happens when the participants disagree, a policy changes, evidence expires, or the world produces an unwelcome result. Can another participant tell what was established, what’s still uncertain, what actions remain permitted, and what must happen next? The reliability of that relationship, not the readability of the model's private state, would be the harder achievement.

The earlier work leading into this research program didn’t start with latent reasoning. It began with questions about selecting and placing computation, diagnosing capabilities that fail to appear when needed, preserving obligations across transient participants, and deciding when evidence authorizes action. In chip design, another strand asked how faster candidate generation changes the importance of evaluators and accumulated engineering memory. Hybrid reasoning draws these strands together because it makes the separation between internal computation and external interaction more explicit [1]. These earlier papers don’t demonstrate a completed architecture. They offer different starting points for testing the integrity of consequential crossings and the economic value of getting them right.

The program’s eventual architecture remains open. The earlier papers provide design distinctions and testable wagers. One possible system uses a control plane to allocate computation; interfaces that preserve distinctions appropriate to each recipient; independent evaluators that establish what evidence supports; an authority mechanism rooted in actual roles and rules; and durable memory that carries conditions and obligations beyond the life of any model instance. The hard parts lie at the boundaries between these functions. Good prediction doesn’t establish a fact; a valid measurement doesn’t confer permission; permission doesn’t

establish a successful effect; an accurate record doesn’t guarantee that its contents remain valid tomorrow. The architecture succeeds only if these separations remain usable under pressure, not just elegant in a diagram.

Natural language may become one exceptionally flexible interface among many, not the compulsory execution medium for every intelligent component. That would make it easier to place different kinds of computation where they work best. It would also expose an old temptation: believing that a coherent explanation is proof of correct interpretation, justified belief, or legitimate authority. There may be no universal crossing record that settles those matters for all recipients. The research challenge is to specify what a particular recipient must distinguish, test whether the interface preserves those distinctions, and change or refuse the interface when it can’t.

A machine may choose its internal representation. The world shouldn’t be required to accept the machine's account of itself as the sole evidence for what its actions mean or whether those actions are permitted. Between the hidden computation and an external consequence, a dependable system must establish what was meant, what the evidence warrants, what the rules authorize, and what obligations survive. These are different achievements. Making these achievements individually testable, and making their crossings capable of recognizing failure, is the research program.

AI Assistance and Editorial Responsibility

This report was developed by Brad Pierce with substantial assistance from an AI model. He established the research objectives and editorial direction and retains responsibility for the claims, source attributions, and decisions about publication.

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